Instructor Guide: Build an OrbitCo System Diagram

Type: Hands-on lab (individual work) **Time:** 15 minutes (12 min individual work + 3 min critique on the projector)

Pre-Activity Setup

- Read the scenario aloud before starting the timer. The five-node flow (L1 → L2 → L3 → Director + Post-mortem Owner as parallel handoff) needs to be clear in every learner's head before they open PowerPoint — without that clarity, the hand-built version cannot communicate the difference between escalation and handoff.
- Have a reference version of both slides on your own laptop. A SmartArt Basic Process with all five nodes, and a hand-built version with the Post-mortem Owner offset. You will project yours during the critique, alongside a learner's.
- Confirm every learner has desktop PowerPoint. The full shapes menu, connector tool, and Arrange → Align are all best accessed from the desktop client.
- If the class uses a shared theme or template (for example, a company-branded master), have the path ready so learners can apply the theme and keep both slide versions visually consistent.

Facilitation Steps

1. **0:00 — Launch** (2 min): Project the scenario. Walk through the five nodes and name the escalation-vs-handoff distinction explicitly: "The Post-mortem Owner is not step 5 in the escalation — it is a parallel handoff that happens after resolution. Your diagram should show that." Start the timer.

2. **0:02 — Circulation, first 3 minutes** (focus: steps 1-2): Watch for learners who skip the paper sketch and go straight into SmartArt. Stop them — the sketch is one minute and forces the flow-direction decision that SmartArt will not ask about. Also watch for learners typing into SmartArt without opening the SmartArt text pane; the pane is the faster input path.

3. **0:05 — Circulation, middle 4 minutes** (focus: steps 3-5): This is where most of the teaching happens. Watch for free-drawn lines (not connectors) — they look right until the learner moves a shape and the line stays put. Fix early. Also watch

for the Post-mortem Owner being drawn identically to the escalation chain — the whole point of the hand-built version is that it can represent the parallel-handoff shape that SmartArt cannot. Push back gently: "What is different about this node conceptually? Does your drawing show that?"

4. **0:09 — Circulation, last 3 minutes** (focus: steps 6-8): Labels on connectors (step 5) and the chosen-final defense (step 7). Unlabeled arrows are the most common weakness in every diagramming lab — catch them in this last window. The chosen-final sentence is also load-bearing: a learner who can not defend the choice has not actually made it.

5. **0:12 — Critique on the projector** (3 min): Project two strong examples side-by-side — one SmartArt, one hand-built. Do a brief public critique. What works? What would you change? The class produces better diagrams the next time they have to draw one because they have seen good ones named specifically on a projector. Close the lab with the context-dependence point: runbook → hand-built; executive slide → SmartArt or simple hand-built.

What to Circulate For

- **Free-drawn lines instead of connectors.** The most common mechanical error. Have the learner grab a shape and drag it around the slide; if the "connector" stays put, it is a free line and needs to be redrawn using **Insert → Shapes → Lines → Connector**. Connectors have small dots on the ends when you hover over a shape; free lines do not.
- **Post-mortem Owner drawn identically to the escalation chain.** The entire argument for the hand-built version is that it can represent a parallel handoff. If the learner drew five identical rectangles in a row, the hand-built version does not actually demonstrate anything SmartArt cannot. Push for a different shape, a different color, or a different position.
- **Unlabeled connectors.** Every arrow should answer "what is happening here?" in one or two words. "Pages," "escalates to," "hands off to" — anything that names the action. Arrows without labels are guessable at best.
- **Inconsistent flow direction.** If the SmartArt is horizontal but the hand-built version is vertical (or vice versa), the class cannot easily compare them. Both

should go the same direction, and that direction should have been decided on paper at step 1.

- **Skipping Arrange → Align.** Four rectangles that are almost aligned feel sloppy. Two clicks fix them. Watch for it specifically in the last three minutes — if the diagram is off by a few pixels, the fix is fast.

Common Pitfalls

- **SmartArt style change re-colors the whole diagram.** Learners sometimes change the color scheme on SmartArt and lose the customization they were about to make. The SmartArt Design → Change Colors dropdown is the right place; SmartArt Design → SmartArt Styles changes the visual effect (raised, inset, cartoon), not the color. Keep Change Colors for palette; Styles for embellishment (and avoid the 3D styles).
- **Connector snap-points are not visible.** On some PowerPoint versions, the green connection dots only appear when you hover directly over a shape edge. If a learner cannot figure out how to attach a connector, have them hover slowly over the shape edge with the connector tool selected.
- **Grouping too early or too late.** If a learner groups the diagram before alignment is final, they cannot adjust individual shapes. If they never group, one accidental click can drag a shape out of place. The right sequence is: align, label connectors, then group.
- **Chosen-final defense is cosmetic rather than structural.** "SmartArt is cleaner" or "the hand-built one has more shapes" is not a defense. The right defense names a tradeoff: speed vs precision, generic vs specific, reusable vs bespoke. Push for the real reason.
- **Saving as `.pptx` (not `.ppt`).** The older format is the default on some installations. Not a huge deal for this lab but worth correcting as a habit — the `.pptx` format preserves theme colors and SmartArt more reliably.

Debrief Prompts

- **"Which version did you pick, and why?"** The answer should cite the tradeoff (SmartArt is fast and consistent; hand-built is precise). If a learner picked SmartArt

"because it looked better," probe — usually they mean "it was faster," and "looked better" is a reflex rather than a real preference.

- **"If this diagram were going into a runbook, which version would you pick? What about for an executive slide?"** Draws out the context-dependence. Runbook: hand-built, with maximum precision. Executive slide: SmartArt or simple hand-built, optimized for speed of reading. Same diagram; different consumer; different choice.
- **"What would you change if the diagram had to cover failure-path branches as well as the happy path?"** Leads to a discussion of when flowcharts with decision diamonds are warranted over a linear process flow. If time allows, sketch a two-branch version on the whiteboard — this previews the runbook-flow work that lands in Lesson 4's capstone preparation.

Expected Answers

A reference end-state for the lab:

Element	Expected value
SmartArt slide	Basic Process template with five nodes (L1, L2, L3, Director, Post-mortem Owner), recolored to the deck's accent palette
Hand-built slide	Four aligned rectangles in a row (L1 → L2 → L3 → Director), a fifth differentiated shape (Post-mortem Owner) offset below or to the side
Connector labels	Every connector has a text label ("pages," "escalates to," "hands off to," or similar)
Alignment	All shapes in the escalation row share the same top coordinate; horizontal spacing is even
Grouping	Each slide's diagram is grouped (Ctrl+G / Cmd+G) so it moves as one unit
Chosen final	A text-box note on at least one slide, naming the choice (SmartArt or hand-built) and a one-sentence defense that references the speed-vs-precision tradeoff

If a learner produced a diagram that breaks one of these expectations but defends the choice (for example, "I chose all rectangles even for the Post-mortem Owner because our team's notation standard does not distinguish handoffs from escalations by shape"), accept it. The goal is defensible notation, not uniform notation.